

GoGet: A multi-model multimodal assistive agent with Disagreement-Gated Grounding

Siddhant Shrivastava, A Manicka Praveen, Malika Meghjani

{siddhant_shrivastava@mymail, praveen_amanicka@mymail, malika_meghjani}@sutd.edu.sg

Singapore University of Technology and Design, Singapore

Abstract—GoGet is a locally hosted multimodal AI agent. It composes four open transformer-family models on one NVIDIA RTX 5090 with 32 GB of VRAM and coordinates them through a shared textual working memory. The four models are Parakeet TDT 0.6B for speech recognition, Gemma-4 as the vision-language reasoner and controller, SAM 3.1 for open-vocabulary segmentation, and Kokoro-82M for speech synthesis. On each turn the agent perceives, routes the request across five operating modes, invokes the matching model or tool, observes the result, and re-plans. Our main contribution is Disagreement-Aware Grounded Fusion (DAGF), a selective-prediction policy for visual counting that treats the segmenter’s count as authoritative only when the segmenter and the reasoner corroborate, and abstains with a hedged answer when they disagree. We evaluate on the live system. With exact ground truth, grounding raises counting accuracy from 0.37 to 0.91; on 260 real COCO images the gain is smaller (0.30 to 0.37), since counting real clutter is hard for both models. DAGF does not improve the count itself; it decides when to trust it. Cross-model agreement predicts a correct count (0.70 when the models agree, 0.23 when they disagree), and abstaining on disagreement more than halves confidently-wrong counts, from 0.63 to 0.25 ($p < 0.001$); the agent then returns a confident correct count on a third of queries and abstains on the rest. A cross-validated confidence floor (DAGF-R) cuts confident errors further on moderate counts (0.26 to 0.14). The intent router reaches a macro-F1 of 1.00, against 0.85 for a keyword baseline. We also embody the agent on a simulated mobile robot. Grounded counting is viewpoint-dependent: a far view is unreliable (0.30 exact-match) while a near view is exact (1.00), so the agent reads DAGF’s abstention as a cue to approach before it reports. Every model is open and runs on a private GPU, so no audio, imagery, or screen content leaves the machine.

Index Terms—multimodal fusion networks, vision-language grounding, selective prediction, explainable neural networks, attention mechanisms, edge inference

I. INTRODUCTION

Transformer models now read speech, images, and text well enough to power assistive AI agents that listen, look, and act from one natural-language interface. A user can ask “how many cups are on the table?”, “open a terminal and print the date”, or “go to the whiteboard and tell me what is on it” within a single session. Two pressures shape how such an agent should be built. The first is privacy. Capable systems are usually served as one large model behind a cloud API. That sends camera and microphone data off the device and rules out on-premises use where privacy matters. The second is trust. A monolithic model that answers “how many cups?” straight from pixels states fine-grained facts, such as object counts, that an assistive user has no way to check.

GoGet runs an agent loop over four compact open models on one NVIDIA RTX 5090 with 32 GB of VRAM. Each transformers-based model has a distinct job. Parakeet TDT 0.6B, a token-and-duration transducer, transcribes speech. Gemma-4 [1], a decoder-only vision-language model, is the agent’s controller: it routes intent, reasons over the grounded scene, selects tools, and plans. SAM 3.1 [2], a Vision Transformer encoder with a promptable mask decoder, performs open-vocabulary segmentation. Kokoro-82M synthesizes speech. A single orchestrator passes context between them as text. The output of one attention-based model conditions the next, and the whole chain fits the latency and memory budget of one GPU and the response time of an interactive agent supporting voice, text, visuals, and actions.

A composable multi-model agent in operation systematically exposes the biases of individual models that a monolithic model can hide due to its black-box nature. Two or more of the agent’s models can *disagree*, and that disagreement is information. We use it to ground the results with more confidence. Our central contribution, Disagreement-Aware Grounded Fusion (DAGF), turns the segmenter and the reasoner into mutual referees for visual counting. Grounding the reasoner on the segmenter’s count works well on clear scenes. Open-vocabulary segmentation miscounts on dense and occluded scenes, and when it does it is *confidently* wrong: every mask scores high while the total is off. Asserting that count anyway is the worst behaviour for an assistive tool. DAGF instead emits a confident count only when the two models corroborate, and abstains with a hedged answer when they diverge. It is a selective-prediction policy over the composed output.

Our contributions are:

- **DAGF**, a selective-prediction policy that calibrates segmentation-grounded counts using cross-model agreement, with a tolerance that traces a risk-coverage operating curve, and a robust two-signal variant (DAGF-R) that also gates on the segmenter’s own confidence.
- **A single-GPU agent** that composes four open transformer models on one RTX 5090, sharing context through a textual working memory and time-multiplexing the GPU so the full stack is co-resident.
- **A measured evaluation on the live system**: grounding lifts counting exact-match from 0.37 to 0.91 with exact ground truth; on 260 real COCO images with gold counts, agreement predicts grounded correctness (0.70 vs 0.23) and DAGF more than halves confident counting errors

($p < 0.001$); the VLM router reaches macro-F1 1.00 against a 0.85 keyword baseline.

- An **embodied extension** in which DAGF’s abstention drives active perception: a simulated robot approaches to resolve a contested count, turning an unreliable far reading (0.30) into a correct near one (1.00).

II. RELATED WORK

Transformer multimodal models. Vision-language models such as Gemma-4 [1] extend decoder-only transformers with image tokens and cross-modal attention; embodied variants such as RT-2 [3] and PaLM-E [4] fold perception and control into one large policy. GoGet keeps perception, reasoning, and action in separate compact models and studies how to share and reconcile context between them on local hardware. **Segmentation transformers.** The Segment Anything family [2], [5], [6] couples a Vision Transformer image encoder with a promptable transformer mask decoder for text-driven masks. GoGet uses these masks as countable evidence and also models their *reliability*. **Selective prediction and uncertainty.** Abstaining when a model is likely wrong is classical [7], usually driven by a single model’s softmax or calibrated confidence [8]. DAGF draws its abstention signal from *disagreement between two pretrained models*, in the lineage of query-by-committee [9] and deep-ensemble disagreement [10]. Its committee is two heterogeneous model types [11], a segmenter and a vision-language reasoner. **Agentic and edge systems.** GUI agents such as OS-Atlas [12] dedicate a model to screen grounding; we reuse one reasoner to conserve GPU memory, and rely on lightweight front ends [13], [14] and ROS 2 for a fully local stack. Across these lines, no prior work treats cross-model disagreement inside a deployed multimodal agent as a calibration signal for a downstream task. Selective prediction and calibration methods [7], [8] gate on a single model’s own confidence; composed agent and GUI-agent systems [12] route between models but do not use inter-model disagreement to decide when a joint output should be trusted. DAGF addresses this gap by treating agreement between a segmenter and a reasoner as the trust signal itself, rather than relying on either model’s internal confidence alone.

III. GOGET ARCHITECTURE AND GROUNDING

A. Architecture

GoGet runs as a staged agent loop inside one FastAPI WebSocket process (Fig. 1). A browser-based frontend (to be showcased in the video demo) streams microphone, camera, and text from a human user. A turn proceeds as: transcription, intent routing, optional perception or tool use, grounded response, and synthesis, with a replan step when an action fails. The four models load once on the RTX 5090 (using about 29 GB of its 32 GB VRAM). The router emits a noun phrase for the segmenter, whose instance count returns to the responder as a grounded constraint; DAGF gates how authoritative that constraint is. The router also dispatches desktop-control and robot turns.

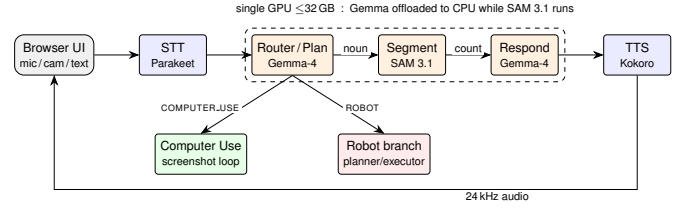


Fig. 1. GoGet interactive turn loop. The dashed region denotes GPU co-residency, with Gemma-4 offloaded to CPU while SAM 3.1 runs.

Text as shared working memory. The models communicate primarily through text. Parakeet emits a transcript based on the user’s speech. Gemma-4 reads the transcript text with the current frame and produces a routing decision and a perception query. SAM 3.1 takes the query and returns a count. Gemma-4 reads the count and composes a reply. Kokoro renders the reply as audio. Text is the common representation that links four models with different input modalities into one chain. It also makes intermediate state legible in the interface as a live trace.

Conversation-aware routing. One Gemma-4 pass over the frame and utterance returns a single label: a perception noun phrase, `COMPUTER_USE`, `ROBOT`, or `NONE`. We supply the last five dialogue turns as context. A follow-up such as “count them again” then inherits the prior target through attention over that history. We also write the per-turn count back into the record, so the router stays consistent as the scene moves.

Single-GPU co-residency. Holding a vision-language model and a Vision Transformer segmenter active at once would exceed 32 GB (the VRAM size of our GPU). Therefore, GoGet serializes the heavy stages: an asynchronous lock keeps the real-time tracking loop from overlapping a full turn, and the reasoner is paged to CPU for each segmentation call and restored afterward. Conversational turns avoid this paging while grounded perception with two large models necessitates it. (§IV). **Operating modes.** The same stack serves five modes through one interface: live-camera question answering, media analysis (video, image, GIF, PDF, or document), live commentary for captioning, tool-using desktop control behind an explicit flag, and robot assistance, where a Gemma-4 planner emits a JSON action plan for a MuJoCo or ROS 2 / Nav2 robot. Because every model is open and self-hosted, audio, frames, and screen content are processed and discarded on the device, with no third-party inference dependency. The architectural novelty here is not any individual model but the combination of a shared textual working memory, conversation-aware routing, and single-GPU co-residency that lets four heterogeneous, independently-trained models operate as one coherent interactive agent within a consumer GPU’s memory budget.

B. Disagreement-Aware Grounded Fusion

Unlike confidence-based selective prediction, which gates on a single model’s own calibrated uncertainty [7], [8], DAGF gates on *inter-model* agreement between two independent

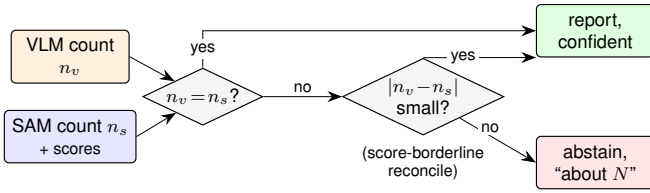


Fig. 2. DAGF decision flow: report a confident count on agreement, abstain with a hedged estimate on disagreement.

models of different types (an image segmenter and a vision-language reasoner) treating their disagreement itself as the reliability signal. Each counting turn (“how many X ?”) is answered by two models. The router emits a noun phrase; SAM 3.1 returns per-instance masks with probabilities and a count n_s ; Gemma-4, prompted to count the same noun, returns an independent count n_v . Naive grounding injects n_s into the responder prompt as authoritative. DAGF (Fig. 2, Eq. 1) instead decides *how* authoritative n_s is.

DAGF reports a confident count when the segmenter and the reasoner corroborate, exactly or within a tolerance, and *abstains* with a hedged “about N ” when they diverge. Concretely, the system *commits* to a count when it returns n_s to the user as a confident, unhedged answer; when it does not commit, it abstains, returning a hedged midpoint estimate flagged as uncertain.

The design follows a fact we measure in §IV: the segmenter’s counting errors carry *high* per-detection scores, so its own confidence cannot flag them, while the segmenter and the reasoner *disagree* far more often on exactly the scenes where the grounded count is wrong. DAGF therefore treats cross-model disagreement as the reliability signal. It commits the grounded count when the two counts agree or fall within an absolute or relative tolerance (δ_a, δ_r); otherwise it abstains, returning a hedged midpoint flagged uncertain so the responder says “about N ”. The tolerance is the knob of a selective predictor: tightening it commits fewer but safer counts, which traces a risk–coverage curve (§IV). With relative gap $\rho = |n_v - n_s| / \max(n_v, n_s)$, DAGF adds one VLM counting pass and no new parameters:

$$\hat{n} = \begin{cases} n_s, & n_v = n_s \quad (\text{commit}) \\ n_s, & |n_v - n_s| \leq \delta_a \text{ or } \rho \leq \delta_r \quad (\text{commit}) \\ \text{round}\left(\frac{n_v + n_s}{2}\right), & \text{otherwise (abstain).} \end{cases} \quad (1)$$

A robust two-signal variant, **DAGF-R**, adds a segmenter-confidence floor: a count that Eq. 1 would commit is downgraded to an abstention when the segmenter’s weakest accepted detection scores below τ_{floor} . This catches counts on which the two models agree yet the segmenter is underconfident (§IV).

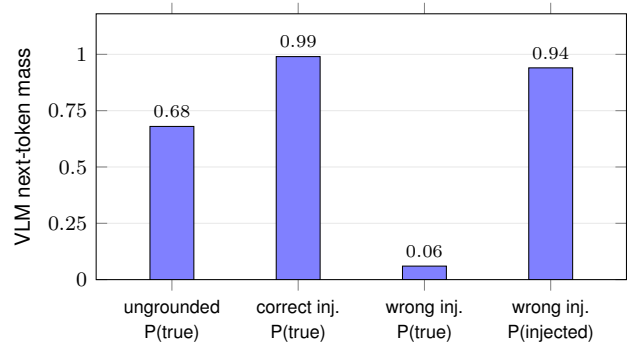


Fig. 3. Gemma-4’s next-token distribution over digit counts, with and without an injected count, correct and counterfactual

IV. EXPERIMENTS

All numbers are measured on the live stack on one RTX 5090.¹ We fix the DAGF tolerances before evaluation ($\delta_a=1$, $\delta_r=0.15$) and sweep them for the risk–coverage curve.

A. Counting and grounding

We first measure grounding with *exact* ground truth. The benchmark is 43 synthetic scenes whose target count is fixed by construction. It spans a difficulty gradient, from sparse and large objects to dense, tiny, and occluded ones. Grounding the reasoner on the segmenter raises exact-match accuracy from 0.37 to 0.91 and cuts mean absolute error from 1.63 to 0.12 (Table I, Fig. 4(a)); the gain is significant (McNemar $p < 0.001$, 23 images corrected and none regressed). The segmenter stays reliable even under high density. It errs only by one at the extreme, and those errors carry high per-detection scores. No borderline detections occurred in any scene, so the segmenter’s own confidence cannot flag its mistakes.

B. How grounding reshapes the reasoner

Grounding works by editing the reasoner’s output distribution, and we measure that edit directly. For each scene we read Gemma-4’s next-token distribution over the digits for “how many X ?”, with and without the count injected (Fig. 3, single-token counts). Injecting the correct count sharpens the distribution onto the truth: mean probability on the true count rises from 0.68 to 0.99 and the distribution’s entropy collapses from 0.56 to 0.03 nats. The attention over one injected token, not a softer prompt edit, is doing the work. A counterfactual makes the risk explicit: when we inject a deliberately *wrong* count, the reasoner adopts it with probability 0.94 and the true count is suppressed to 0.06. The reasoner repeats whatever the segmenter asserts, right or wrong. This is precisely why a composed agent must *gate* grounding: unconditional trust propagates the segmenter’s error.

¹Exact identifiers, for reproducibility: google/gemma-4-E4B-it; the SAM 3.1 *multiplex* checkpoint; Parakeet TDT 0.6B v2; Kokoro-82M. Our evaluation harness and the synthetic and natural counting benchmarks are released with the code.

TABLE I
COUNTING: REASONER-ONLY VS GROUNDED VS DAGF AND ITS ROBUST
VARIANT DAGF-R (SEGMENTER-CONFIDENCE FLOOR).

Set	Method	Exact $\uparrow$	Conf-err $\downarrow$	Coverage
Synthetic (exact GT, n=43)	VLM-only	0.37	0.63	1.00
	Grounded	0.91	0.09	1.00
	DAGF	0.79	0.07	0.86
COCO real (gold, n=260)	VLM-only	0.30	0.70	1.00
	Grounded	0.37	0.63	1.00
	DAGF	0.37	0.25	0.57
COCO counts ≤ 10 (n=160)	Grounded	0.54	0.46	1.00
	DAGF	–	0.26	0.73
	DAGF-R	–	0.14	0.51

C. Disagreement-aware fusion on real images

We validate DAGF on 260 real COCO val images with gold instance counts (counts 2–25). Grounding raises exact-match from 0.30 to 0.37, a significant gain on real images (McNemar $p = 0.03$), though counting real clutter stays hard for both models. DAGF does not push exact-match above grounding (both 0.37); its job is to decide when the count is trustworthy. The abstention signal holds on real data: the grounded count is correct in 0.70 of the images where the two models agree, against only 0.23 where they disagree. DAGF reads this and more than halves the confidently-wrong rate, from 0.63 (95% Wilson CI [0.57, 0.68]) to 0.25 ([0.20, 0.30]) (exact McNemar $p < 0.001$, 99 images fixed, none worsened). Concretely, of all queries the agent then returns a confident correct count on 32%, abstains on 43%, and is confidently wrong on 25%, down from the 63% confidently-wrong of blanket grounding. The agreement signal is strongest where the segmenter actually errs: the accuracy gap between agreeing and disagreeing scenes is wide on COCO (0.70 vs 0.23) and on the generated-natural set (0.55 vs 0.14), but narrow on the synthetic shapes (0.94 vs 0.88), where the segmenter is almost always right and there is little to flag.

High counts. The set spans an easy and a hard regime. On counts ≤ 10 (n=160) grounding reaches 0.54 and DAGF answers 73% of images at 0.26 confident error. On counts 11–25 (n=100) counting is hard for both models (grounding 0.10), and DAGF responds by abstaining on 70% of images while still cutting confident error from 0.90 to 0.23. The gate degrades gracefully: where neither model can count, it mostly declines to assert one.

DAGF-R: a segmenter-confidence floor. Some errors survive because both models agree on a wrong count, and on COCO these coincide with a low minimum detection score. DAGF-R adds a second gate: it abstains when the segmenter’s weakest accepted detection scores below a floor, even when the two counts agree. We select the floor by 5-fold cross-validation (it lands stably at ≈ 0.7) and score it on held-out folds, so it is never tuned on the data it is measured on. On the moderate regime, where coverage headroom exists, DAGF-R lowers confident error from 0.26 to 0.14 and raises committed precision from 0.65 to 0.73, at 0.51 coverage (Table I). Two

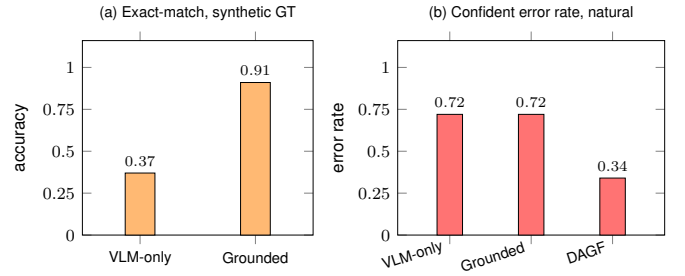


Fig. 4. (a) Counting accuracy with and without grounding, exact ground truth. (b) Confidently-wrong rate with and without DAGF, natural cluttered scenes.

TABLE II
ABLATION OF THE DAGF GATE ON POOLED EXACT-GT COUNTS.

Gate variant	Coverage	Committed error $\downarrow$
No gate (naive grounding)	1.00	0.24
Agreement-only ($\delta=0$)	0.24	0.08
DAGF default ($\delta_a=1, \delta_r=0.15$)	0.68	0.19
Loose ($\delta_a=2, \delta_r=0.25$)	0.94	0.25

richer alternatives do not help: a learned logistic gate over disagreement and score features does not beat this simple two-signal rule, and replacing the single-shot reasoner with a self-consistency median over five samples leaves the numbers unchanged. DAGF therefore stays simple.

D. Ablations and selective prediction

Sweeping the tolerance turns DAGF into a selective predictor with a tunable operating point (Table II, pooled exact-GT counts, $n=103$, base error 0.24). Committing only on exact agreement is safest, at 0.08 committed error, but it answers 24% of counts; the default tolerance answers 68% at 0.19 error; a loose tolerance answers 94% with error back at the no-gate level, so the gate helps only while it stays selective. Ranking counts by cross-model agreement and committing the most-agreed first reaches an area-under-risk-coverage of 0.13 against 0.17 for random ordering. A logistic calibrator fit to predict grounded correctness puts almost all of its weight on the relative disagreement, confirming that disagreement, not count magnitude, carries the signal. The single-model baseline, the segmenter’s own score, is regime-limited: it ranks errors on clean synthetic shapes yet stays flat on natural clutter, where only disagreement separates correct counts from incorrect ones.

E. Routing and latency

We test the router on 120 balanced utterances across perception, desktop, and chat. It reaches accuracy and macro-F1 of 1.00 with no cross-class confusion. A keyword-matching baseline reaches 0.85, but it recovers only 0.70 of desktop commands and over-routes to chat (precision 0.72 on the chat class), which is the failure mode the VLM router removes. The set is broad but lexically separable, so the ceiling result mainly shows the router avoids the baseline’s class confusions; harder

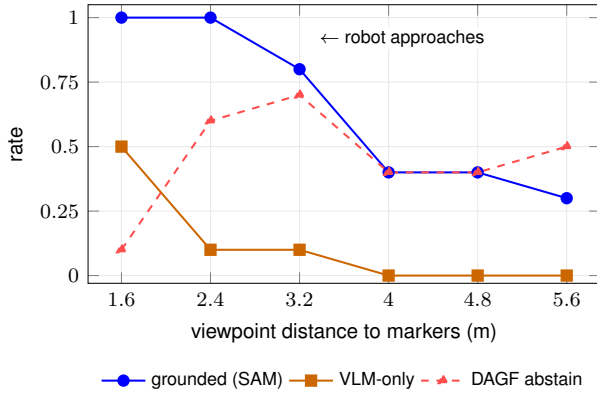


Fig. 5. Rate (grounded counting accuracy rate and DAGF abstention rate) vs. standoff distance from markers, MuJoCo, exact ground truth. Robot Approaches the designated markers from right to left.

pronoun and ambiguous utterances are future work. We also time the loop. Conversational turns finish at a 0.31 s median (p95 0.32 s), with routing and response at 0.15 s and 0.11 s. Grounded-perception turns take 4.75 s (p95 4.81 s), almost all in the 4.49 s segmentation stage, which is mostly the cost of paging Gemma-4 to CPU and back for co-residency. For the speech front end, a prior controlled benchmark selected Parakeet TDT 0.6B (word error rate 0.116 at 0.27 s, the best of the candidates including Whisper [15]) and Kokoro for its eight-language coverage at a comparable cost.

F. Embodied grounding: counting under object approach

The robot mode lets us ask what the agent should do when DAGF abstains: a mobile base can move to see better. In MuJoCo with exact ground truth, a robot views N red markers from six standoff distances (10 scenes, $N=4$ to 8). Grounded counting is strongly viewpoint-dependent: from 5.6 m the markers merge and accuracy is 0.30; it reaches 1.00 by 2.4 m, while the reasoner alone stays near zero (Fig. 5). DAGF’s abstention falls with proximity (0.50 far to 0.10 near), a “move closer” cue. The closed loop is only partly solved: committing at the first confident pose reaches 0.40, because at long range both models undercount together and agree on a wrong count; re-approaching whenever the far view is contested reaches 0.70, against 1.00 for always approaching to a near standoff. The gate flags far views as uncertain, but only a near view resolves them.

G. Discussion

When disagreement helps. DAGF assumes both models are good enough to referee each other. That assumption breaks on the synthetic set. There the reasoner saturates near 24 on high counts, so it disagrees with a *correct* segmenter and triggers false abstentions. This is why DAGF’s synthetic exact-match (0.79) sits below naive grounding (0.91). The policy is built for the natural regime, where both models are fallible and disagreement is informative. A count-aware tolerance and a stronger counting reasoner would narrow the gap.

Scope and other limitations. We evaluate the gate on visual counting only; extending disagreement-gating to presence, attributes, or spatial relations is open. COCO gold counts include occluded and tiny instances, which caps measurable accuracy and partly explains the modest grounding gain on real images. The embodied gate also depends on a competent referee: adding coloured distractors to the MuJoCo scene made the reasoner undercount the target at every distance, so DAGF over-abstained, which is why we report the homogeneous scene. Finally, grounding quality follows segmentation quality, and grounded-perception latency is a few seconds because co-residency pages the reasoner on each turn.

Toward embodied deployment. The MuJoCo robot matches the differential-drive kinematics of a Clearpath Jackal [16], and the same plan grammar targets Nav2 [17]. The active-perception policy [18] should therefore transfer to a real UGV, with the front camera as the frame source.

Safety and privacy. The desktop control mode runs shell commands and is therefore kept opt-in via a launch-based flag. Robot trials use an emergency stop and a geofence. All inference is local and open-weight, so no sensor data leaves the device, and the orchestrator logs every routed action for audit.

V. CONCLUSION

GoGet composes four open transformer models into one assistive agent on a single consumer GPU and reconciles their outputs through a shared working memory. Its central mechanism, DAGF, treats the disagreement between a segmenter and a vision-language reasoner as a calibration signal: it asserts a grounded count only when the two corroborate and abstains otherwise. On synthetic scenes grounding raises counting accuracy to 0.91; on real images, where counting is hard for both models, DAGF leaves accuracy unchanged but halves the rate of confidently-wrong counts by abstaining on disagreement. Treating cross-model disagreement as information is what lets a composed local agent be honest about what it does not know.

ACKNOWLEDGMENT

This work is supported by the TL@SUTD Seed Research Programme, which encourages innovative research in data and information exploitation. We also thank the developers of SAM3.1 (Meta), Gemma4 (Google), Parakeet (Nvidia), Kokoro, and Nav2, whose open models and tooling made this system and study possible.

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